Lab 3 - Assignment: Clean the NYC Taxi data, then stop

Course: INF4058 Data Engineering | Due: 19th of October 11:59pm| Submit: One PDF report + GitHub repo link

Goal

In class you ran the pipeline end to end: PostgreSQL in Docker, ingest_data.py loading 1,369,765 rows, and the same script packaged as a container. At home you look at the data, find its problems, decide what to do with each one, and write the cleaning step. You stop at cleaning.

Do not load the cleaned data into Postgres yet: next session we connect it to the pipeline and run everything together.

What You Do (Steps)

Step	What you do	Evidence in your report
1. Look	Start Jupyter and read the first 100 rows of yellow_tripdata_2021-01.csv.gz. What is a row? Which columns are dates? Which look empty?	Printed: df.head() and df.dtypes
2. Profile	Read the whole file in chunks of 100,000 rows. Never load it all at once. Do not modify the data.	Printed table: dtype, nulls (count and %), min and max per column
3. Find problems	Look for pickups outside January 2021, negative or zero fares and distances, passenger_count = 0, duplicate rows. Write each one down.	Printed tables with the counts behind each problem
4. Decide 5. Clean	For each problem choose: drop, fix or flag. Say why, and who should own that rule. Write clean(df). It returns the cleaned dataframe AND a dict {rule: rows_removed}. Never drop rows silently; do not change the original dataframe.	A short table: problem, decision, why. Printed: rows before, rows after, rows removed per rule
6. Stop here	Keep the code in a notebook or a .py file. Do not load cleaned data into Postgres.	Your repo with the notebook or script (no data, no .venv)
7. Report	Write the report described below and push your repo.	One PDF + the repo link

The Report (One PDF)

Section	What to write	Evidence
1. Cover	Name, ID, date, repo link	
2. The flow, in your words	Source → Extract → Transform → Load → Serve. What ingest_data.py, docker-compose.yaml and the Dockerfile each do, and why (chunks, Int64, replace on the first chunk).	A small diagram or one paragraph per stage
3. What we ran in class	Compose up ingest COUNT = 1,369,765 docker build container run into yellow_taxi_trips. One sentence each.	Screenshots taken in class: the command AND its output
4. What you found in the data	Nulls, bad dates, odd values, duplicates, with the numbers.	Printed tables: print(df) of the profiling results
5. How you cleaned it	Each rule: drop, fix or flag, and why. Rows removed per rule, before and after.	Printed table: {rule: rows_removed} and the counts
6. Answers	Q1: Why does the container reach the DB as pgdatabase, not localhost? Q2: Why chunks, and what was wrong with replace for many months? Q3: How did you verify your results?	
7. What broke	One error you hit, its cause, and how you fixed it.	Screenshots: the error and the fix

Rules

- **Screenshots:** Take them during class. Each one shows the command, its output and your terminal prompt. You do not repeat the class steps at home.
- **Tables:** Print every table with print() (for example print(df)) and screenshot the output. No typed or cropped tables.
- **Your own words:** Explain the flow and your decisions yourself. You may use any tool, including an AI assistant, to help you write or debug code, but you must understand every line and be able to explain it **(I will ask in the lab)**.
- **Repo:** Push your notebook or script. Do not push data files, the .venv folder or any keys.
- **Stop at cleaning:** Do not load the cleaned data into Postgres. We connect it to the pipeline next session.